

Retentive Network 公式推导

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1 注意事项

- 若不同小节中的 notation 含义不一致，以当前小节的为准
- 小写形式的矩阵会加粗
- 灰色框内的是公式推导部分

2 Forward

给定输入序列: $x_1, x_2, \dots, x_n$

$$X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ \dots \\ X_n \end{bmatrix} \in R^{n \times d_{model}}$$

将每个时间步的向量 X_n 乘以权重矩阵 W_V 映射为一维向量 v_n :

$$v_n = X_n \cdot W_V$$

其中 $X_n \in R^{1 \times d}$, $W_V \in R^{d \times d}$, $v_n \in R^{1 \times d}$

2.1 Retention 公式 (v_n to o_n)

$$s_n = A s_{n-1} + K_n^T v_n \tag{1}$$

$$o_n = Q_n s_n = \sum_{m=1}^n Q_n A^{n-m} K_m^T v_m \tag{2}$$

其中 $A \in R^{d \times d}$, $K_n \in R^{1 \times d}$, $s_n \in R^{d \times d}$, $Q_n \in R^{1 \times d}$

推导：

$$\begin{aligned}
s_n &= A s_{n-1} + K_n^\top v_n \\
&= A(A s_{n-2} + K_{n-1}^\top v_{n-1}) + K_n^\top v_n \\
&= A^2 s_{n-2} + A K_{n-1}^\top v_{n-1} + K_n^\top v_n \\
&= A^3 s_{n-3} + A^2 K_{n-2}^\top v_{n-2} + A K_{n-1}^\top v_{n-1} + K_n^\top v_n \\
&= \dots \\
&= A^{n-1} K_1^\top v_1 + \dots + A^3 K_{n-3}^\top v_{n-3} + A^2 K_{n-2}^\top v_{n-2} + A K_{n-1}^\top v_{n-1} + K_n^\top v_n \\
&= \sum_{m=1}^n A^{n-m} K_m^\top v_m
\end{aligned}$$

接下来，我们通过下面的公式得到 Q_n, K_n ：

$$Q = XW_Q, \quad K = XW_K \quad (3)$$

其中 $W_Q, W_K \in R^{d \times d}$ 是可以学习的参数矩阵。

由于几乎任何的矩阵都可以在复数域上相似对角化，故得 $A = \Lambda(\gamma e^{i\theta})\Lambda^{-1}$, where $\gamma, \theta \in R^d$. 根据对角阵的性质得 $A^{n-m} = \Lambda(\gamma e^{i\theta})^{n-m}\Lambda^{-1}$. 将 Λ 和 Λ^{-1} 分别吸收进 W_Q 和 W_K ，可以将公式 (2) 重写为：

$$\begin{aligned}
o_n &= \sum_{m=1}^n Q_n(\gamma e^{i\theta})^{n-m} K_m^\top v_m \\
&= \sum_{m=1}^n (Q_n(\gamma e^{i\theta})^n)(K_m(\gamma e^{i\theta})^{-m})^\top v_m
\end{aligned} \quad (4)$$

进一步简化 γ 为一个标量，则公式 (3) 变为：

$$o_n = \sum_{m=1}^n \gamma^{n-m} (Q_n e^{in\theta})(K_m e^{im\theta})^\dagger v_m \quad (5)$$

其中 $\dagger$ 代表共轭转置，并且有：

$$\begin{aligned}
e^{in\theta} &= \text{diag}(e^{in\theta_1}, e^{in\theta_2}, \dots, e^{in\theta_d}) \in R^{d \times d} \\
e^{im\theta} &= \text{diag}(e^{im\theta_1}, e^{im\theta_2}, \dots, e^{im\theta_d}) \in R^{d \times d}
\end{aligned}$$

推导: $\gamma, \theta \in R^d$, 即 $\gamma = [\gamma_1, \gamma_2, \dots, \gamma_d], \theta = [\theta_1, \theta_2, \dots, \theta_d]$

$$\gamma e^{i\theta} = \begin{bmatrix} \gamma_1 e^{i\theta_1} & & & \\ & \gamma_2 e^{i\theta_2} & & \\ & & \dots & \\ & & & \gamma_d e^{i\theta_d} \end{bmatrix} \in R^{d \times d}$$

$$A = \Lambda(\gamma e^{i\theta})\Lambda^{-1} = \Lambda \begin{bmatrix} \gamma_1 e^{i\theta_1} & & & \\ & \gamma_2 e^{i\theta_2} & & \\ & & \dots & \\ & & & \gamma_d e^{i\theta_d} \end{bmatrix} \Lambda^{-1}$$

$$A^{n-m} = (\Lambda(\gamma e^{i\theta})\Lambda^{-1})^{n-m} = \Lambda(\gamma e^{i\theta})^{n-m}\Lambda^{-1}$$

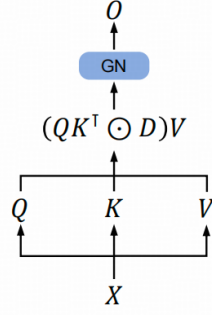
$$\begin{aligned} o_n &= \sum_{m=1}^n Q_n A^{n-m} K_m^\top v_m \\ &= \sum_{m=1}^n Q_n \Lambda(\gamma e^{i\theta})^{n-m} \Lambda^{-1} K_m^\top v_m \\ &= \sum_{m=1}^n Q_n (\gamma e^{i\theta})^{n-m} K_m^\top v_m \quad (Q_n = Q_n \Lambda, K_m^\top = \Lambda^{-1} K_m^\top) \\ &= \sum_{m=1}^n \gamma^{n-m} Q_n (e^{i\theta})^{n-m} K_m^\top v_m \quad (\text{Simplify } \gamma \text{ to scalar}) \\ &= \sum_{m=1}^n \gamma^{n-m} (Q_n e^{in\theta}) (e^{-im\theta} K_m^\top) v_m \\ &= \sum_{m=1}^n \gamma^{n-m} (Q_n e^{in\theta}) (K_m e^{im\theta})^\dagger v_m \quad (\dagger \text{ is the conjugate transpose}) \end{aligned}$$

2.2 The Parallel Representation of Retention

Retention 的并行表示如 Figure 1所示:

$$\begin{aligned} Q &= (XW_Q) \odot \Theta, \quad K = (XW_K) \odot \overline{\Theta}, \quad V = XW_V \\ \Theta_n &= [e^{in\theta_1} \quad e^{in\theta_2} \quad \dots \quad e^{in\theta_d}] \in R^{1 \times d} \\ \overline{\Theta}_n &= [e^{-in\theta_1} \quad e^{-in\theta_2} \quad \dots \quad e^{-in\theta_d}] \in R^{1 \times d} \\ D_{nm} &= \begin{cases} \gamma^{n-m} & n \geq m \\ 0 & n < m \end{cases} \\ \text{Retention}(X) &= (QK^\top \odot D) \end{aligned} \tag{6}$$

其中 $Q \in R^{n \times d}, K \in R^{n \times d}, V \in R^{n \times d}, \Theta \in R^{n \times d}$



(a) Parallel representation.

Figure 1: Parallel representation.

推导:

由公式 (5):

$$\begin{aligned}
 \text{Retention}(X) &= \begin{bmatrix} o_1 \\ o_2 \\ \dots \\ o_n \end{bmatrix} = \begin{bmatrix} \sum_{m=1}^1 \gamma^{1-m} (Q_1 e^{i\theta}) (K_m e^{im\theta})^\dagger v_m \\ \sum_{m=1}^2 \gamma^{2-m} (Q_2 e^{i2\theta}) (K_m e^{im\theta})^\dagger v_m \\ \dots \\ \sum_{m=1}^n \gamma^{n-m} (Q_n e^{in\theta}) (K_m e^{im\theta})^\dagger v_m \end{bmatrix} = \begin{bmatrix} \sum_{m=1}^1 \gamma^{1-m} (Q_1 e^{i\theta}) (K_m e^{-im\theta})^\top v_m \\ \sum_{m=1}^2 \gamma^{2-m} (Q_2 e^{i2\theta}) (K_m e^{-im\theta})^\top v_m \\ \dots \\ \sum_{m=1}^n \gamma^{n-m} (Q_n e^{in\theta}) (K_m e^{-im\theta})^\top v_m \end{bmatrix} \\
 &= \begin{bmatrix} (Q_1 e^{i\theta}) (K_1 e^{-i\theta})^\top & 0 & \dots & 0 \\ \gamma (Q_2 e^{i2\theta}) (K_1 e^{-i\theta})^\top & (Q_2 e^{i2\theta}) (K_2 e^{-i2\theta})^\top & \dots & 0 \\ \dots & \dots & \dots & 0 \\ \gamma^{n-1} (Q_n e^{in\theta}) (K_1 e^{-i\theta})^\top & \gamma^{n-2} (Q_n e^{in\theta}) (K_n e^{-in\theta})^\top & \dots & (Q_n e^{in\theta}) (K_n e^{-in\theta})^\top \end{bmatrix} @ \begin{bmatrix} v_1 \\ v_2 \\ \dots \\ v_n \end{bmatrix} \\
 &= \left(\begin{bmatrix} (Q_1 e^{i\theta}) (K_1 e^{-i\theta})^\top & (Q_1 e^{i\theta}) (K_2 e^{-i2\theta})^\top & \dots & (Q_1 e^{i\theta}) (K_n e^{-in\theta})^\top \\ (Q_2 e^{i2\theta}) (K_1 e^{-i\theta})^\top & (Q_2 e^{i2\theta}) (K_2 e^{-i2\theta})^\top & \dots & (Q_2 e^{i2\theta}) (K_n e^{-in\theta})^\top \\ \dots & \dots & \dots & \dots \\ (Q_n e^{in\theta}) (K_1 e^{-i\theta})^\top & (Q_n e^{in\theta}) (K_2 e^{-i2\theta})^\top & \dots & (Q_n e^{in\theta}) (K_n e^{-in\theta})^\top \end{bmatrix} \odot \begin{bmatrix} 1 & 0 & \dots & 0 \\ \gamma & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ \gamma^{n-1} & \gamma^{n-2} & \dots & 1 \end{bmatrix} \right) @ \begin{bmatrix} v_1 \\ v_2 \\ \dots \\ v_n \end{bmatrix} \\
 &= \left(\begin{bmatrix} Q_1 e^{i\theta} \\ Q_2 e^{i2\theta} \\ \dots \\ Q_n e^{in\theta} \end{bmatrix} @ \begin{bmatrix} (K_1 e^{-i\theta})^\top & (K_2 e^{-i2\theta})^\top & \dots & (K_n e^{-in\theta})^\top \end{bmatrix} \right) \odot \begin{bmatrix} 1 & 0 & \dots & 0 \\ \gamma & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ \gamma^{n-1} & \gamma^{n-2} & \dots & 1 \end{bmatrix} @ \begin{bmatrix} v_1 \\ v_2 \\ \dots \\ v_n \end{bmatrix} \\
 &= \left(\left(\begin{bmatrix} Q_1 e^{i\theta} \\ Q_2 e^{i2\theta} \\ \dots \\ Q_n e^{in\theta} \end{bmatrix} @ \begin{bmatrix} K_1 e^{-i\theta} \\ K_2 e^{-i2\theta} \\ \dots \\ K_n e^{-in\theta} \end{bmatrix} \right) \odot \begin{bmatrix} 1 & 0 & \dots & 0 \\ \gamma & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ \gamma^{n-1} & \gamma^{n-2} & \dots & 1 \end{bmatrix} \right) @ \begin{bmatrix} v_1 \\ v_2 \\ \dots \\ v_n \end{bmatrix} \\
 &= \left(\left(\begin{bmatrix} Q_1 \\ Q_2 \\ \dots \\ Q_n \end{bmatrix} \odot \begin{bmatrix} \Theta_1 \\ \Theta_2 \\ \dots \\ \Theta_n \end{bmatrix} \right) @ \begin{bmatrix} K_1 \\ K_2 \\ \dots \\ K_n \end{bmatrix} \odot \begin{bmatrix} \overline{\Theta_1} \\ \overline{\Theta_2} \\ \dots \\ \overline{\Theta_n} \end{bmatrix} \right) \odot \begin{bmatrix} 1 & 0 & \dots & 0 \\ \gamma & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ \gamma^{n-1} & \gamma^{n-2} & \dots & 1 \end{bmatrix} @ \begin{bmatrix} v_1 \\ v_2 \\ \dots \\ v_n \end{bmatrix} \\
 &= (((XW_Q) \odot \Theta) @ ((XW_K) \odot \overline{\Theta})^\top) \odot D @ (XW_V) \\
 &= (QK^\top \odot D)V
 \end{aligned}$$

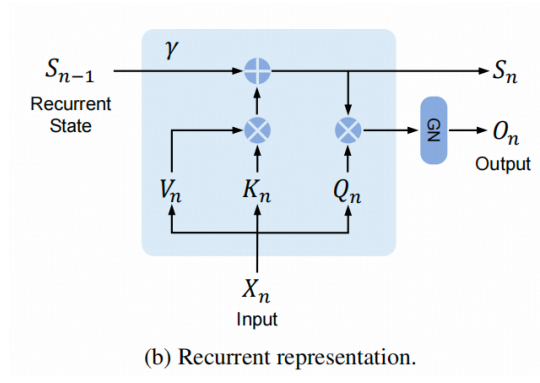


Figure 2: Recurrent representation.

2.3 The Recurrent Representation of Retention

Retention 的回归表示如 Figure 2 所示. 对于第 n 个时间步, 我们可以在推理时回归地得到每个输出向量:

$$\begin{aligned} S_n &= \gamma S_{n-1} + K_n^\top V_n \\ \text{Retention}(X_n) &= Q_n S_n \end{aligned} \quad (7)$$

其中 $Q = (XW_Q) \odot \Theta$, $K = (XW_K) \odot \bar{\Theta}$, $V = (XW_V)$

推导:

由公式 (6) 得:

$$\begin{aligned} \text{Retention}(X) &= (QK^\top \odot D)V \\ &= \left(\begin{bmatrix} Q_1 \\ Q_2 \\ \dots \\ Q_n \end{bmatrix} @ \begin{bmatrix} K_1^\top, K_2^\top, \dots, K_n^\top \end{bmatrix} \right) \odot \begin{bmatrix} 1 & 0 & \dots & 0 \\ \gamma & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ \gamma^{n-1} & \gamma^{n-2} & \dots & 1 \end{bmatrix} @ \begin{bmatrix} V_1 \\ V_2 \\ \dots \\ V_n \end{bmatrix} \\ &= \left(\begin{bmatrix} Q_1 K_1^\top & Q_1 K_2^\top & \dots & Q_1 K_n^\top \\ Q_2 K_1^\top & Q_2 K_2^\top & \dots & Q_2 K_n^\top \\ \dots & \dots & \dots & \dots \\ Q_n K_1^\top & Q_n K_2^\top & \dots & Q_n K_n^\top \end{bmatrix} \odot \begin{bmatrix} 1 & 0 & \dots & 0 \\ \gamma & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ \gamma^{n-1} & \gamma^{n-2} & \dots & 1 \end{bmatrix} \right) @ \begin{bmatrix} V_1 \\ V_2 \\ \dots \\ V_n \end{bmatrix} \\ &= \begin{bmatrix} Q_1 K_1^\top & 0 & \dots & 0 \\ \gamma Q_2 K_1^\top & Q_2 K_2^\top & \dots & 0 \\ \dots & \dots & \dots & \dots \\ \gamma^{n-1} Q_n K_1^\top & \gamma^{n-2} Q_n K_2^\top & \dots & Q_n K_n^\top \end{bmatrix} @ \begin{bmatrix} V_1 \\ V_2 \\ \dots \\ V_n \end{bmatrix} \end{aligned}$$

$$\begin{aligned}
&= \begin{bmatrix} \sum_{m=1}^1 \gamma^{1-m} Q_1 K_m^\top V_m \\ \sum_{m=1}^2 \gamma^{2-m} Q_2 K_m^\top V_m \\ \dots \\ \sum_{m=1}^n \gamma^{n-m} Q_n K_m^\top V_m \end{bmatrix} \\
&= \begin{bmatrix} \text{Retention}(X_1) \\ \text{Retention}(X_2) \\ \dots \\ \text{Retention}(X_n) \end{bmatrix}
\end{aligned}$$

即：

$$\begin{aligned}
\text{Retention}(X_n) &= \sum_{m=1}^n \gamma^{n-m} Q_n K_m^\top V_m \\
&= Q_n \sum_{m=1}^n \gamma^{n-m} K_m^\top V_m \\
&= Q_n S_n \quad (\text{let } S_n = \sum_{m=1}^n \gamma^{n-m} K_m^\top V_m)
\end{aligned}$$

则：

$$\begin{aligned}
S_n &= \sum_{m=1}^{n-1} \gamma^{n-m} K_m^\top V_m + K_n^\top V_n \\
&= \gamma \sum_{m=1}^{n-1} \gamma^{(n-1)-m} K_m^\top V_m + K_n^\top V_n \\
&= \gamma S_{n-1} + K_n^\top V_n
\end{aligned}$$

2.4 The Chunkwise Recurrent Representation of Retention

$$\begin{aligned}
Q_{[i]} &= Q_{[B(i-1)+1:Bi]}, \quad K_{[i]} = K_{[B(i-1)+1:Bi]}, \quad V_{[i]} = V_{[B(i-1)+1:Bi]} \\
R_i &= K_{[i]}^\top (V_{[i]} \odot \zeta) + \gamma^B R_{i-1}, \quad \zeta_{ij} = \gamma^{B-i-1} \\
\text{Retention}(X_{[i]}) &= \underbrace{(Q_{[i]} K_{[i]}^\top \odot D) V_{[i]}}_{\text{Inner-Chunk}} + \underbrace{Q_{[i]} R_{i-1} \odot \xi}_{\text{Cross-Chunk}}, \quad \xi_{ij} = \gamma^{i+1}
\end{aligned} \tag{8}$$

其中：

$$\zeta_{ij} = \gamma^{B-i-1}, i.e. \zeta = \begin{bmatrix} \gamma^{B-1} & \gamma^{B-1} & \dots & \gamma^{B-1} \\ \gamma^{B-2} & \gamma^{B-2} & \dots & \gamma^{B-2} \\ \dots & \dots & \dots & \dots \\ 1 & 1 & \dots & 1 \end{bmatrix}_{B \times d}$$

$$\xi_{ij} = \gamma^{i+1}, i.e. \xi = \begin{bmatrix} \gamma & \gamma & \dots & \gamma \\ \gamma^2 & \gamma^2 & \dots & \gamma^2 \\ \dots & \dots & \dots & \dots \\ \gamma^B & \gamma^B & \dots & \gamma^B \end{bmatrix}_{B \times d}$$

$$D_{nm} = \begin{cases} \gamma^{n-m} & n \geq m \\ 0 & n < m \end{cases}, i.e. D = \begin{bmatrix} 1 & 0 & \dots & 0 \\ \gamma & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ \gamma^{B-1} & \gamma^{B-2} & \dots & 1 \end{bmatrix}_{B \times B}$$

推导：

$$Retention(X_n) = \sum_{m=1}^n \gamma^{n-m} Q_n K_m^\top V_m$$

$$Retention(X_{[i]}) = \begin{bmatrix} Retention(X_{(i-1)B+1}) \\ Retention(X_{(i-1)B+2}) \\ \dots \\ Retention(X_{iB}) \end{bmatrix} = \begin{bmatrix} \sum_{m=1}^{(i-1)B+1} \gamma^{(i-1)B+1-m} Q_{(i-1)B+1} K_m^\top V_m \\ \sum_{m=1}^{(i-1)B+2} \gamma^{(i-1)B+2-m} Q_{(i-1)B+2} K_m^\top V_m \\ \dots \\ \sum_{m=1}^{iB} \gamma^{iB-m} Q_{iB} K_m^\top V_m \end{bmatrix}$$

$$= \underbrace{\begin{bmatrix} \sum_{m=(i-1)B+1}^{(i-1)B+1} \gamma^{(i-1)B+1-m} Q_{(i-1)B+1} K_m^\top V_m \\ \sum_{m=(i-1)B+1}^{(i-1)B+2} \gamma^{(i-1)B+2-m} Q_{(i-1)B+2} K_m^\top V_m \\ \dots \\ \sum_{m=(i-1)B+1}^{iB} \gamma^{iB-m} Q_{iB} K_m^\top V_m \end{bmatrix}}_{Retention(X_{[i]})_{Inner-Chunk}} + \underbrace{\begin{bmatrix} \sum_{m=1}^{(i-1)B} \gamma^{(i-1)B+1-m} Q_{(i-1)B+1} K_m^\top V_m \\ \sum_{m=1}^{(i-1)B} \gamma^{(i-1)B+2-m} Q_{(i-1)B+2} K_m^\top V_m \\ \dots \\ \sum_{m=1}^{(i-1)B} \gamma^{iB-m} Q_{iB} K_m^\top V_m \end{bmatrix}}_{Retention(X_{[i]})_{Cross-Chunk}}$$

$$Retention(X_{[i]})_{Inner-Chunk} = \begin{bmatrix} \sum_{m=(i-1)B+1}^{(i-1)B+1} \gamma^{(i-1)B+1-m} Q_{(i-1)B+1} K_m^\top V_m \\ \sum_{m=(i-1)B+1}^{(i-1)B+2} \gamma^{(i-1)B+2-m} Q_{(i-1)B+2} K_m^\top V_m \\ \dots \\ \sum_{m=(i-1)B+1}^{iB} \gamma^{iB-m} Q_{iB} K_m^\top V_m \end{bmatrix}$$

$$= \begin{bmatrix} Q_{(i-1)B+1} K_{(i-1)B+1}^\top & 0 & \dots & 0 \\ \gamma Q_{(i-1)B+2} K_{(i-1)B+1}^\top & Q_{(i-1)B+2} K_{(i-1)B+2}^\top & \dots & 0 \\ \dots & \dots & \dots & \dots \\ \gamma^{B-1} Q_{iB} K_{(i-1)B+1}^\top & \gamma^{B-2} Q_{iB} K_{(i-1)B+2}^\top & \dots & Q_{iB} K_{iB}^\top \end{bmatrix} @ \begin{bmatrix} V_{(i-1)B+1} \\ V_{(i-1)B+2} \\ \dots \\ V_{iB} \end{bmatrix}$$

$$\begin{aligned}
&= \left(\begin{bmatrix} Q_{(i-1)B+1} K_{(i-1)B+1}^\top & Q_{(i-1)B+1} K_{(i-1)B+2}^\top & \dots & Q_{(i-1)B+1} K_{iB}^\top \\ Q_{(i-1)B+2} K_{(i-1)B+1}^\top & Q_{(i-1)B+2} K_{(i-1)B+2}^\top & \dots & Q_{(i-1)B+2} K_{iB}^\top \\ \dots & \dots & \dots & \dots \\ Q_{iB} K_{(i-1)B+1}^\top & Q_{iB} K_{(i-1)B+2}^\top & \dots & Q_{iB} K_{iB}^\top \end{bmatrix} \odot \begin{bmatrix} 1 & 0 & \dots & 0 \\ \gamma & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ \gamma^{B-1} & \gamma^{B-2} & \dots & 1 \end{bmatrix} \right) @ \begin{bmatrix} V_{(i-1)B+1} \\ V_{(i-1)B+2} \\ \dots \\ V_{iB} \end{bmatrix} \\
&= \left(\left(\begin{bmatrix} Q_{(i-1)B+1} \\ Q_{(i-1)B+2} \\ \dots \\ Q_{iB} \end{bmatrix} @ \begin{bmatrix} K_{(i-1)B+1}^\top & K_{(i-1)B+2}^\top & \dots & K_{iB}^\top \end{bmatrix} \right) \odot \begin{bmatrix} 1 & 0 & \dots & 0 \\ \gamma & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ \gamma^{B-1} & \gamma^{B-2} & \dots & 1 \end{bmatrix} \right) @ \begin{bmatrix} V_{(i-1)B+1} \\ V_{(i-1)B+2} \\ \dots \\ V_{iB} \end{bmatrix} \\
&= (Q_{[i]} K_{[i]}^\top \odot D) V_{[i]} \\
&= \text{Retention}(X_{[i]})_{\text{Cross-Chunk}} = \begin{bmatrix} \sum_{m=1}^{(i-1)B} \gamma^{(i-1)B+1-m} Q_{(i-1)B+1} K_m^\top V_m \\ \sum_{m=1}^{(i-1)B} \gamma^{(i-1)B+2-m} Q_{(i-1)B+2} K_m^\top V_m \\ \dots \\ \sum_{m=1}^{(i-1)B} \gamma^{iB-m} Q_{iB} K_m^\top V_m \end{bmatrix} \\
&= \begin{bmatrix} Q_{(i-1)B+1} \sum_{m=1}^{(i-1)B} \gamma^{(i-1)B+1-m} K_m^\top V_m \\ Q_{(i-1)B+2} \sum_{m=1}^{(i-1)B} \gamma^{(i-1)B+2-m} K_m^\top V_m \\ \dots \\ Q_{iB} \sum_{m=1}^{(i-1)B} \gamma^{iB-m} K_m^\top V_m \end{bmatrix} = \begin{bmatrix} \gamma Q_{(i-1)B+1} \sum_{m=1}^{(i-1)B} \gamma^{(i-1)B-m} K_m^\top V_m \\ \gamma^2 Q_{(i-1)B+2} \sum_{m=1}^{(i-1)B} \gamma^{(i-1)B-m} K_m^\top V_m \\ \dots \\ \gamma^B Q_{iB} \sum_{m=1}^{(i-1)B} \gamma^{(i-1)B-m} K_m^\top V_m \end{bmatrix} \\
&= \begin{bmatrix} \gamma Q_{(i-1)B+1} \\ \gamma^2 Q_{(i-1)B+2} \\ \dots \\ \gamma^B Q_{iB} \end{bmatrix}_{B \times d} @ \left(\sum_{m=1}^{(i-1)B} \gamma^{(i-1)B-m} K_m^\top V_m \right)_{d \times d} \\
&= \left(\begin{bmatrix} \gamma & \gamma & \dots & \gamma \\ \gamma^2 & \gamma^2 & \dots & \gamma^2 \\ \dots & \dots & \dots & \dots \\ \gamma^B & \gamma^B & \dots & \gamma^B \end{bmatrix}_{B \times d} \odot \begin{bmatrix} Q_{(i-1)B+1} \\ Q_{(i-1)B+2} \\ \dots \\ Q_{iB} \end{bmatrix}_{B \times d} \right) @ \left(\sum_{m=1}^{(i-1)B} \gamma^{(i-1)B-m} K_m^\top V_m \right)_{d \times d} \\
&= (\xi \odot Q_{[i]}) @ \left(\sum_{m=1}^{(i-1)B} \gamma^{(i-1)B-m} K_m^\top V_m \right)_{d \times d} \\
&= (\xi \odot Q_{[i]}) @ R_{i-1} \quad (\text{let } R_{i-1} = \sum_{m=1}^{(i-1)B} \gamma^{(i-1)B-m} K_m^\top V_m) \\
&= Q_{[i]} R_{i-1} \odot \xi \quad (\text{Because every row in } \xi \text{ is the same}) \\
&R_i = \sum_{m=1}^{iB} \gamma^{iB-m} K_m^\top V_m = \sum_{m=1}^{(i-1)B} \gamma^{iB-m} K_m^\top V_m + \sum_{m=(i-1)B+1}^{iB} \gamma^{iB-m} K_m^\top V_m \\
&= \gamma^B \sum_{m=1}^{(i-1)B} \gamma^{(i-1)B-m} K_m^\top V_m + (\gamma^{B-1} K_{(i-1)B+1}^\top V_{(i-1)B+1} + \gamma^{B-2} K_{(i-1)B+2}^\top V_{(i-1)B+2} + \dots + K_{iB}^\top V_{iB})
\end{aligned}$$

$$\begin{aligned}
&= \gamma^B R_{i-1} + \begin{bmatrix} K_{(i-1)B+1}^\top & K_{(i-1)B+2}^\top & \dots & K_{iB}^\top \end{bmatrix}_{d \times B} @ \begin{bmatrix} \gamma^{B-1} V_{(i-1)B+1} \\ \gamma^{B-2} V_{(i-1)B+2} \\ \dots \\ V_{iB} \end{bmatrix}_{B \times d} \\
&= \gamma^B R_{i-1} + \begin{bmatrix} K_{(i-1)B+1}^\top & K_{(i-1)B+2}^\top & \dots & K_{iB}^\top \end{bmatrix}_{d \times B} @ \left(\begin{bmatrix} V_{(i-1)B+1} \\ V_{(i-1)B+2} \\ \dots \\ V_{iB} \end{bmatrix}_{B \times d} \odot \begin{bmatrix} \gamma^{B-1} & \gamma^{B-1} & \dots & \gamma^{B-1} \\ \gamma^{B-2} & \gamma^{B-2} & \dots & \gamma^{B-2} \\ \dots & \dots & \dots & \dots \\ 1 & 1 & \dots & 1 \end{bmatrix}_{B \times d} \right) \\
&= \gamma^B R_{i-1} + K_{[i]}^\top (V_{[i]} \odot \zeta)
\end{aligned}$$

3 Backward

根据 Chunkwise 公式：

$$\begin{aligned}
R_i &= \gamma^B R_{i-1} + K_{[i]}^\top (V_{[i]} \odot \zeta) \in R^{d \times d} \\
O_{[i]} &= \text{Retention}(X_{[i]}) = (Q_{[i]} K_{[i]}^\top \odot D) V_{[i]} + Q_{[i]} R_{i-1} \odot \xi \in R^{B \times d}
\end{aligned}$$

其中：

$$\begin{aligned}
\zeta_{ij} &= \gamma^{B-i-1}, i.e. \zeta = \begin{bmatrix} \gamma^{B-1} & \gamma^{B-1} & \dots & \gamma^{B-1} \\ \gamma^{B-2} & \gamma^{B-2} & \dots & \gamma^{B-2} \\ \dots & \dots & \dots & \dots \\ 1 & 1 & \dots & 1 \end{bmatrix}_{B \times d} \\
\xi_{ij} &= \gamma^{i+1}, i.e. \xi = \begin{bmatrix} \gamma & \gamma & \dots & \gamma \\ \gamma^2 & \gamma^2 & \dots & \gamma^2 \\ \dots & \dots & \dots & \dots \\ \gamma^B & \gamma^B & \dots & \gamma^B \end{bmatrix}_{B \times d} \\
D_{nm} &= \begin{cases} \gamma^{n-m} & n \geq m \\ 0 & n < m \end{cases}, i.e. D = \begin{bmatrix} 1 & 0 & \dots & 0 \\ \gamma & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ \gamma^{B-1} & \gamma^{B-2} & \dots & 1 \end{bmatrix}_{B \times B}
\end{aligned}$$

则：

$$dR_{i-1} = \gamma^B dR_i + Q_{[i]}^\top (dO_{[i]} \odot \xi) \quad (9)$$

$$dQ_{[i]} = (dO_{[i]} \odot \xi) R_{i-1}^\top + ((dO_{[i]} V_{[i]}^\top) \odot D) K_{[i]} \quad (10)$$

$$dK_{[i]} = (V_{[i]} \odot \zeta) dR_i^\top + ((dO_{[i]} V_{[i]}^\top) \odot D)^\top Q_{[i]} \quad (11)$$

$$dV_{[i]} = (K_{[i]} dR_i) \odot \zeta + ((Q_{[i]} K_{[i]}^\top) \odot D)^\top dO_{[i]} \quad (12)$$

推导：

由：

$$\begin{aligned} R_i &= \gamma^B R_{i-1} + K_{[i]}^\top \\ dR_{i-1}^{(1)} &= \gamma^B dR_i \\ dK_{[i]}^{(1)} &= (V_{[i]} \odot \zeta) dR_i^\top \\ dV_{[i]}^{(1)} &= (K_{[i]} dR_i) \odot \zeta \end{aligned}$$

由：

$$O_{[i]} = (Q_{[i]} K_{[i]}^\top \odot D) V_{[i]} + \underline{Q_{[i]} R_{i-1} \odot \xi}$$

可得：

$$\begin{aligned} dQ_{[i]}^{[1]} &= (dO_{[i]} \odot \xi) R_{i-1}^\top \\ dR_{i-1}^{[2]} &= Q_{[i]}^\top (dO_{[i]} \odot \xi) \end{aligned}$$

由：

$$O_{[i]} = \underline{(Q_{[i]} K_{[i]}^\top \odot D) V_{[i]}} + Q_{[i]} R_{i-1} \odot \xi$$

可得：

$$\begin{aligned} dV_{[i]}^{[2]} &= ((Q_{[i]} K_{[i]}^\top) \odot D)^\top dO_{[i]} \\ dQ_{[i]}^{[2]} &= ((dO_{[i]} V_{[i]}^\top) \odot D) K_{[i]} \\ dK_{[i]}^{[2]} &= ((dO_{[i]} V_{[i]}^\top) \odot D)^\top Q_{[i]} \end{aligned}$$

故：

$$\begin{aligned} dR_{i-1} &= dR_{i-1}^{[1]} + dR_{i-1}^{[2]} = \gamma^B dR_i + Q_{[i]}^\top (dO_{[i]} \odot \xi) \\ dQ_{[i]} &= dQ_{[i]}^{[1]} + dQ_{[i]}^{[2]} = (dO_{[i]} \odot \xi) R_{i-1}^\top + ((dO_{[i]} V_{[i]}^\top) \odot D) K_{[i]} \\ dK_{[i]} &= dK_{[i]}^{[1]} + dK_{[i]}^{[2]} = (V_{[i]} \odot \zeta) dR_i^\top + ((dO_{[i]} V_{[i]}^\top) \odot D)^\top Q_{[i]} \\ dV_{[i]} &= dV_{[i]}^{[1]} + dV_{[i]}^{[2]} = (K_{[i]} dR_i) \odot \zeta + ((Q_{[i]} K_{[i]}^\top) \odot D)^\top dO_{[i]} \end{aligned}$$

4 Algorithm

Forward 公式 (Chunkwise 版本):

$$\begin{aligned}
 Q_{[i]} &= Q_{[B(i-1)+1:Bi]}, \quad K_{[i]} = K_{[B(i-1)+1:Bi]}, \quad V_{[i]} = V_{[B(i-1)+1:Bi]} \\
 R_i &= K_{[i]}^\top (V_{[i]} \odot \zeta) + \gamma^B R_{i-1}, \quad \zeta_{ij} = \gamma^{B-i-1} \\
 \text{Retention}(X_{[i]}) &= \underbrace{(Q_{[i]} K_{[i]}^\top \odot D) V_{[i]}}_{\text{Inner-Chunk}} + \underbrace{Q_{[i]} R_{i-1} \odot \xi}_{\text{Cross-Chunk}}, \quad \xi_{ij} = \gamma^{i+1}
 \end{aligned}$$

Algorithm 1: Forward

Input: $Q, K, V \in R^{n \times d}, \gamma$, block size B

Result: $O \in R^{n \times d}$

Divide X into $T = \frac{n}{B}$ blocks, $X_1, X_2, \dots, X_T \in R^{B \times d}, X \in Q, K, V, O$

Initialize $\zeta = \begin{bmatrix} \gamma^{B-1} & \gamma^{B-1} & \dots & \gamma^{B-1} \\ \gamma^{B-2} & \gamma^{B-2} & \dots & \gamma^{B-2} \\ \dots & \dots & \dots & \dots \\ 1 & 1 & \dots & 1 \end{bmatrix}_{B \times d}$

Initialize $\xi = \begin{bmatrix} \gamma & \gamma & \dots & \gamma \\ \gamma^2 & \gamma^2 & \dots & \gamma^2 \\ \dots & \dots & \dots & \dots \\ \gamma^B & \gamma^B & \dots & \gamma^B \end{bmatrix}_{B \times d}$

Initialize $D \in R^{B \times B}, \quad D_{nm} = \begin{cases} \gamma^{n-m} & , n \geq m \\ 0 & , n < m \end{cases}$

Initialize $R = 0 \in R^{d \times d}$

for $t = 1, 2, \dots, T$ **do**

 Load $Q_t, K_t, V_t \in R^{B \times d}$ from HBM to on-chip SRAM

 On chip, compute $O^{[1]} = ((Q_t K_t^\top) \odot D) V_t$

 On chip, compute $O^{[2]} = Q_t R \odot \xi$

 On chip, compute $R = \gamma^B R + K_t^\top (V_t \odot \zeta)$

 Write $O_t = O^{[1]} + O^{[2]}$ to HBM as the t -th block of O

end

return O

Backward 公式：

$$\begin{aligned}
dR_{i-1} &= dR_{i-1}^{[1]} + dR_{i-1}^{[2]} = \gamma^B dR_i + Q_{[i]}^\top (dO_{[i]} \odot \xi) \\
dQ_{[i]} &= dQ_{[i]}^{[1]} + dQ_{[i]}^{[2]} = (dO_{[i]} \odot \xi) R_{i-1}^\top + ((dO_{[i]} V_{[i]}^\top) \odot D) K_{[i]} \\
dK_{[i]} &= dK_{[i]}^{[1]} + dK_{[i]}^{[2]} = (V_{[i]} \odot \zeta) dR_i^\top + ((dO_{[i]} V_{[i]}^\top) \odot D)^\top Q_{[i]} \\
dV_{[i]} &= dV_{[i]}^{[1]} + dV_{[i]}^{[2]} = (K_{[i]} dR_i) \odot \zeta + ((Q_{[i]} K_{[i]}^\top) \odot D)^\top dO_{[i]}
\end{aligned}$$

Algorithm 2: Backward

Input: $Q, K, V, dO \in R^{n \times d}, \gamma$, block size B

Result: $dQ, dK, dV \in R^{n \times d}$

Divide X into $T = \frac{n}{B}$ blocks, $X_1, X_2, \dots, X_T \in R^{B \times d}, X \in Q, K, V$

Divide dX into $T = \frac{n}{B}$ blocks, $dX_1, dX_2, \dots, dX_T \in R^{B \times d}, X \in Q, K, V, O$

$$\text{Initialize } \zeta = \begin{bmatrix} \gamma^{B-1} & \gamma^{B-1} & \dots & \gamma^{B-1} \\ \gamma^{B-2} & \gamma^{B-2} & \dots & \gamma^{B-2} \\ \dots & \dots & \dots & \dots \\ 1 & 1 & \dots & 1 \end{bmatrix}_{B \times d} \quad \text{Initialize } \xi = \begin{bmatrix} \gamma & \gamma & \dots & \gamma \\ \gamma^2 & \gamma^2 & \dots & \gamma^2 \\ \dots & \dots & \dots & \dots \\ \gamma^B & \gamma^B & \dots & \gamma^B \end{bmatrix}_{B \times d}$$

$$\text{Initialize } D \in R^{B \times B}, \quad D_{nm} = \begin{cases} \gamma^{n-m}, & n \geq m \\ 0, & n < m \end{cases}$$

Initialize $R = 0 \in R^{d \times d}, dR = 0 \in R^{d \times d}$

for $t = 1, 2, \dots, T$ **do**

 Load $K_t, V_t, O_t, dO_t \in R^{B \times d}$ from HBM to on-chip SRAM

 On chip, compute $dQ^{[1]} = (dO_t \odot \xi) R^\top$

 On chip, compute $dQ^{[2]} = ((dO_t V_t^\top) \odot D) K_t$

 On chip, compute $R = \gamma^B R + K_t^\top (V_t \odot \zeta)$

 Write $dQ_t = dQ^{[1]} + dQ^{[2]}$ to HBM as the t-th block of Q

end

for $t = T, T-1, \dots, 1$ **do**

 Load $Q_t, K_t, V_t, O_t, dO_t \in R^{B \times d}$ from HBM to on-chip SRAM

 On chip, compute $dK^{[1]} = (V_t \odot \zeta) dR^\top$

 On chip, compute $dK^{[2]} = ((dO_t V_t^\top) \odot D)^\top Q_t$

 On chip, compute $dV^{[1]} = (K_t dR) \odot \zeta$

 On chip, compute $dV^{[2]} = ((Q_t K_t^\top) \odot D)^\top dO_t$

 On chip, compute $dR = \gamma^B dR + Q_t^\top (dO_t \odot \xi)$

 Write $dK_t = dK^{[1]} + dK^{[2]}$ to HBM as the t-th block of dK

 Write $dV_t = dV^{[1]} + dV^{[2]}$ to HBM as the t-th block of dV

end

return dQ, dK, dV
